

A SIMPLE GROUP WITH FULL MF RADICAL

SAUERS

A countable group is MF if it embeds in the unitary group of a norm matrix corona. We construct a nontrivial finitely generated simple Kazhdan group H such that every homomorphism from H to an MF group is trivial. Equivalently, the MF radical of H is all of H , so H is non-MF. We take

$$H = \mathrm{EL}_{16}(L_{\mathbb{F}_2}(1, 2)).$$

Conjugation by an explicit Whitehead matrix maps a Kazhdan subgroup $L \cong \mathrm{EL}_4(L_{\mathbb{F}_2}(1, 2))$ into itself. If c centralizes L , one-sided Kazhdan transport implies that the commutators $[ucu^{-1}, \ell]$ converge to the identity in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation. A finite-order active-core theorem shows that universal Hilbert–Schmidt triviality and MF invisibility coincide for a torsion normal generator. For our choice of u, c , and ℓ , the commutator is an involutory elementary matrix that normally generates H . Every finite-dimensional linear representation of H over any field is also trivial. The same group yields a general construction: for every countable group Q , there is a split epimorphism $\pi_Q: W_Q \rightarrow Q$ such that

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

If Q is MF, the kernel of π_Q is the MF radical of W_Q and is normally generated by the single element d above. For every Q , this kernel is itself a perfect group with full MF radical and admits an explicit decomposition as copies of H amalgamated over $\langle d \rangle$. If every quotient of Q is MF, then the single relation $d = 1$ classifies MF throughout the quotient lattice of W_Q : for every normal subgroup N , W_Q/N is MF exactly when $d \in N$. For $Q = \mathbb{Z}$, the ambient group is finitely generated while its exact MF radical is infinitely generated but normally generated by d .

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1. INTRODUCTION

A countable group is *MF* if it embeds in the unitary group of a norm matrix corona

$$\mathcal{Q}_d = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Equivalently, it admits finite-dimensional unitary models whose multiplicative defects tend to zero in operator norm and which asymptotically separate its nonidentity elements. This is the convention of Carrión–Dadarlat–Eckhardt [CDE].

For a countable group G , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

The group G is MF precisely when $\text{Rad}_{\text{MF}}(G)$ is trivial. If $\text{Rad}_{\text{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \leq G$ define its *MF closure* by

$$\text{cl}_{\text{MF}}^G(N) = \bigcap \{ \ker f : N \leq \ker f, f: G \rightarrow M, M \text{ MF} \}.$$

Thus $\text{Rad}_{\text{MF}}(G) = \text{cl}_{\text{MF}}^G(1)$, and G/N is MF precisely when $\text{cl}_{\text{MF}}^G(N) = N$.

The existence of a countable non-MF group was open [Sh]. Theorem A answers this question and proves the stronger equality $\text{Rad}_{\text{MF}}(H) = H$ for an explicit simple group.

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(1) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem A. *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{16}(R)$. Then H is nontrivial, simple, finitely generated, and has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF.

Call an element of G *universally Hilbert–Schmidt trivial* if its value converges to the identity in normalized Hilbert–Schmidt norm in every operator-norm asymptotic representation of G .

Theorem B (finite-order normal generators). *Let G be finitely generated and let $a \in G$ have finite order and normally generate G . Then*

$$a \in \text{Rad}_{\text{MF}}(G) \iff a \in R_{\infty \rightarrow 2}(G).$$

Consequently, if a is universally Hilbert–Schmidt trivial, then $\text{Rad}_{\text{MF}}(G) = G$.

Suppose that $L \leq G$ has property (T), that $u \in G$ satisfies $uLu^{-1} \leq L$, and that c centralizes L . For every operator-norm asymptotic representation (V_n) of G ,

$$(2) \quad \left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \quad (\ell \in L).$$

Inside H , an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the upper-left subgroup $L \cong \mathrm{EL}_4(R)$. The element $c = e_{45}(1)$ centralizes L , and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

Two elementary commutator identities imply that d normally generates H . Since the coefficient ring has characteristic two, d is an involution. Theorem B then gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Thus property (T) is used for the compressed subgroup L in the transport step; the passage from the Hilbert–Schmidt obstruction to the MF radical uses only the finite order and normal generation of d .

For an exact finite-dimensional representation, conjugation by a compressor defines an injective endomorphism of the commutant of L . This endomorphism is surjective because the commutant is finite-dimensional. Consequently, every finite-dimensional linear representation of H over every field is trivial.

The group H also gives non-MF groups with prescribed MF quotients. The following theorem is proved in Section 7.

Theorem C (prescribed MF quotients). *For every countable group Q , there are a countable group W_Q and a split epimorphism $\pi_Q: W_Q \rightarrow Q$ such that*

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q},$$

where d is the element in Proposition 6.2. Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M . For every $N \trianglelefteq W_Q$, this factorization determines the entire MF closure:

$$\mathrm{cl}_{\mathrm{MF}}^{W_Q}(N) = \pi_Q^{-1}(\mathrm{cl}_{\mathrm{MF}}^Q(\pi_Q(N))).$$

If every quotient of Q is MF, then for every $N \trianglelefteq W_Q$,

$$\mathrm{cl}_{\mathrm{MF}}^{W_Q}(N) = N \langle\langle d \rangle\rangle_{W_Q}, \quad W_Q/N \text{ is MF} \iff d \in N.$$

Writing $K_Q = \ker \pi_Q$, one also has

$$\begin{aligned} \mathrm{Rad}_{\mathrm{MF}}(K_Q) &= K_Q, & [K_Q, K_Q] &= K_Q, \\ Z(K_Q) &= 1, & W_Q &\cong K_Q \rtimes Q. \end{aligned}$$

The element d normally generates K_Q itself. The group K_Q is finitely generated exactly when Q is finite, and its isomorphism type depends only on the cardinality of Q . In particular, one fixed countable perfect group with full MF radical occurs as K_Q for every countably infinite Q .

Relation to prior work. Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations yield the internal endomorphism of the elementary matrix group used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser's normal-subgroup theorem for elementary groups over exchange rings implies that H is simple [Pre]. The theorem of Ershov and Jaikin-Zapirain gives property (T) for elementary groups over finitely generated rings [EJZ]; standard background on property (T) may be found in [BHV]. Carrión, Dadarlat, and Eckhardt relate the MF property for groups to quasidiagonality of group C^* -algebras [CDE], and Shulman studies permanence of the MF property under amalgamated free products [Sh]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT]; Dadarlat studies obstructions to matricial stability in [Dad].

2. ONE-SIDED COMPRESSION IN FINITE DIMENSION

Theorem 2.1 (finite-dimensional commutant rigidity). *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k -vector space, and $\rho: G \rightarrow \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. It commutes with every $\rho(\ell)$, $\ell \in L$, which proves the last assertion. $\square$

Let $\mathrm{Comp}_G(L)$ consist of the $u \in G$ satisfying $uLu^{-1} \leq L$, and let $G^+(L)$ be the subgroup they generate. Define

$$(3) \quad D(G, L) = \langle\langle [gcg^{-1}, \ell] : g \in G^+(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

Corollary 2.2. *Every finite-dimensional linear representation of G , over every field, is trivial on $D(G, L)$. In particular, if $D(G, L) = G$, then every such representation of G is trivial.*

Proof. For a fixed representation ρ , the elements $g \in G$ for which $\rho(g)C\rho(g)^{-1} = C$, where $C = \rho(L)'$, form a subgroup. Theorem 2.1 shows that this subgroup contains $\text{Comp}_G(L)$ and therefore $G^+(L)$. Every commutator in (3) lies in $\ker \rho$, so its normal closure does as well. $\square$

3. KAZHDAN TRANSPORT IN NORMALIZED HILBERT–SCHMIDT NORM

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An operator-norm asymptotic representation of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \rightarrow 0\}.$$

The operator-norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G . Define

$$(4) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where V ranges over all operator-norm asymptotic representations of G . An element is universally Hilbert–Schmidt trivial precisely when it belongs to $R_{\infty \rightarrow 2}(G)$.

Lemma 3.1. *For every sequence (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v . Then $x_n^*x_n \rightarrow 1$ in norm. For all sufficiently large n , the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. In a stably finite unital C^* -algebra, equivalent projections $p \leq q$ satisfy $p = q$. $\square$

Theorem 3.2 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator-norm asymptotic representation of G , and let (x_n) be uniformly bounded in operator norm. If*

$$\|V_n(\ell)x_nV_n(\ell)^* - x_n\|_2 \rightarrow 0 \quad (\ell \in L),$$

then, for every $\ell \in L$,

$$\begin{aligned} \|V_n(\ell)V_n(u)x_nV_n(u)^*V_n(\ell)^* - V_n(u)x_nV_n(u)^*\|_2 &\longrightarrow 0, \\ \|V_n(\ell)V_n(u)^*x_nV_n(u)V_n(\ell)^* - V_n(u)^*x_nV_n(u)\|_2 &\longrightarrow 0. \end{aligned}$$

Proof. Let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the Hilbert-space ultraproduct $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

Its action on $\mathcal{K}_\omega$ is $\sigma(g)$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection of L under $\tilde{\sigma}|_L$. On $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$ [BHV, Chapter 3]. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies

$$U^*PU \leq P.$$

The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$. Hence both U and U^* preserve $\text{Fix } \sigma(L)$, so $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. The two limits in the statement therefore vanish along ω . Since this holds for every free ultrafilter, both sequences converge ordinarily. $\square$

Corollary 3.3. *Let $L \leq G$ have property (T). Then*

$$D(G, L) \leq R_{\infty \rightarrow 2}(G).$$

Proof. Let (V_n) be an operator-norm asymptotic representation. If $c \in C_G(L)$, the sequence $(V_n(c))$ asymptotically commutes with $V_n(L)$ in operator norm and hence in normalized Hilbert–Schmidt norm. Repeated application of Theorem 3.2 shows that the same is true of $(V_n(gcg^{-1}))$ for every $g \in G^+(L)$. Therefore

$$\|V_n([gcg^{-1}, \ell]) - 1\|_2 \longrightarrow 0 \quad (\ell \in L).$$

The subgroup $K_2(V)$ is normal, so it contains the normal closure of these elements. Intersecting over (V_n) proves the claim. $\square$

4. FINITE-ORDER ACTIVE CORES AND THE MF RADICAL

The normalized Hilbert–Schmidt norm can lose a fixed operator-norm defect when that defect occupies a vanishing fraction of the matrix dimension. A finite-order normal generator prevents this loss after an intrinsic reblocking: the active spectral subspace of the generator expands, through fixed normal words, to an invariant subspace whose dimension is only a fixed multiple larger.

Theorem 4.1 (active-core reblocking). *Let G be generated by a finite symmetric set S containing an element a of order $m \geq 2$, and suppose that a normally generates G . For each $s \in S$, fix an expression of s as a product of conjugates of $a^{\pm 1}$, taking the expression for a to be the one-letter word a . Let L be the total number of occurrences of $a^{\pm 1}$ in these expressions.*

If $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_d)$ satisfies $\rho(a) \neq 1$, then, after passing to an infinite coordinate subsequence, ρ has unitary lifts of the form

$$\sigma_n(g) \oplus 1_{d_n - k_n}, \quad \sigma_n(g) \in \mathcal{U}(k_n),$$

where (σ_n) is an operator-norm asymptotic representation. Moreover, $A_n = \sigma_n(a)$ satisfies $A_n^m = 1$, and if

$$p_n = 1_{\mathbb{C} \setminus \{1\}}(A_n), \quad r_n = \text{rank } p_n,$$

then

$$(5) \quad 1 \leq r_n \leq k_n \leq Lr_n.$$

Consequently, with

$$\eta_m = \min\{|1 - \zeta| : \zeta^m = 1, \zeta \neq 1\} > 0,$$

one has

$$(6) \quad \|\sigma_n(a) - 1\|_2 \geq \frac{\eta_m}{\sqrt{L}}.$$

If a is a product of N_g conjugates of $g^{\pm 1}$, then also

$$(7) \quad \liminf_n \|\sigma_n(g) - 1\|_2 \geq \frac{\eta_m}{N_g \sqrt{L}}.$$

Proof. Choose unitary lifts $U_n(g)$ of $\rho(g)$. They form an operator-norm asymptotic representation. Since $\rho(a)^m = 1$, finite-spectrum functional calculus replaces $U_n(a)$, within $o(1)$ in operator norm, by a unitary B_n satisfying $B_n^m = 1$.

Write the chosen expression for $s \in S$ as

$$s = \prod_{j=1}^{L_s} h_{s,j} a^{\varepsilon_{s,j}} h_{s,j}^{-1}, \quad \varepsilon_{s,j} \in \{1, -1\}.$$

Define

$$\tilde{U}_n(s) = \prod_{j=1}^{L_s} U_n(h_{s,j}) B_n^{\varepsilon_{s,j}} U_n(h_{s,j})^*.$$

For every $g \in G$, choose once and for all a word in S representing g and define $\tilde{U}_n(g)$ by substituting these matrices. We use the one-letter word for a , so $\tilde{U}_n(a) = B_n$. The defining group identities and asymptotic multiplicativity give

$$\|\tilde{U}_n(g) - U_n(g)\| \longrightarrow 0 \quad (g \in G).$$

Thus $(\tilde{U}_n)$ is an operator-norm asymptotic representation and represents the same corona homomorphism.

Let P_n be the active spectral subspace of B_n , and let K_n be the sum of the subspaces

$$U_n(h_{s,j})P_n \quad (s \in S, 1 \leq j \leq L_s).$$

Every factor in the displayed formula for $\tilde{U}_n(s)$ is the identity on $K_n^\perp$, preserves K_n , and differs from the identity by an operator whose range lies in K_n . Hence every $\tilde{U}_n(g)$ preserves K_n and is the identity on $K_n^\perp$. Since the one-letter expression for a occurs in the family,

$$\dim P_n \leq \dim K_n \leq L \dim P_n.$$

The assumption $\rho(a) \neq 1$ makes P_n nonzero on an infinite coordinate subsequence. Retain that subsequence, put $k_n = \dim K_n$, and let $\sigma_n(g)$ be the restriction of $\tilde{U}_n(g)$ to K_n . This proves the lifting assertion and (5).

Every nonidentity eigenvalue of A_n is an m th root of unity, so

$$\|A_n - 1\|_2^2 \geq \eta_m^2 \frac{r_n}{k_n} \geq \frac{\eta_m^2}{L},$$

which proves (6). Finally, substitute the matrices $\sigma_n(g)$ into a fixed normal word of length N_g expressing a through $g^{\pm 1}$. Telescoping in normalized Hilbert–Schmidt norm and asymptotic multiplicativity give

$$\|\sigma_n(a) - 1\|_2 \leq N_g \|\sigma_n(g) - 1\|_2 + o(1).$$

Together with (6), this proves (7). $\square$

Proof of Theorem B. For every group,

$$\text{Rad}_{\text{MF}}(G) \leq R_{\infty \rightarrow 2}(G) :$$

an operator-norm asymptotic representation defines a homomorphism to its norm matrix corona, and an element killed there converges to the identity in operator norm, hence in normalized Hilbert–Schmidt norm.

Conversely, suppose $a \in R_{\infty \rightarrow 2}(G)$ but $a \notin \text{Rad}_{\text{MF}}(G)$. Some corona homomorphism sees a . Theorem 4.1 produces an operator-norm asymptotic representation for which (6) holds, contradicting the definition of $R_{\infty \rightarrow 2}(G)$. Thus a lies in the MF radical. Since that radical is normal and a normally generates G , it equals G . $\square$

Corollary 4.2 (simple torsion dichotomy). *Let G be a nontrivial finitely generated simple group containing a finite-order nonidentity element. Then*

$$\text{Rad}_{\text{MF}}(G) = R_{\infty \rightarrow 2}(G).$$

Exactly one of the following holds:

- (1) *every homomorphism from G to a norm matrix corona is trivial;*
- (2) *G admits a faithful operator-norm asymptotic representation (σ_n) such that*

$$\liminf_n \|\sigma_n(g) - 1\|_2 > 0 \quad (g \neq 1).$$

Proof. Both displayed residuals are normal subgroups. Simplicity and Theorem B, applied to a nonidentity torsion element, give their equality. If a corona homomorphism is nontrivial, it is injective. Apply Theorem 4.1; for each $g \neq 1$, simplicity puts the chosen torsion element in the normal closure of g , so (7) gives the asserted pointwise separation. The two alternatives are therefore exhaustive and mutually exclusive. $\square$

5. THE BINARY LEAVITT SELF-COMPRESSION

Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0 t_0, \quad q = s_1 t_1.$$

The relations (1) give

$$(8) \quad p + q = 1, \quad t_1 q s_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 15$ for 16×16 matrices and identify $\mathrm{GL}_4(R)$ with the subgroup of $\mathrm{GL}_{16}(R)$ consisting of the matrices $\mathrm{diag}(A, I_{12})$. On $M_4(R)$ consider the unital injective endomorphism

$$(9) \quad \Psi(A) = qI_4 + s_0 A t_0.$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q s_0 = t_0 q = 0$ and $t_0 s_0 = 1$ show that Ψ is multiplicative and unital. The identity $t_0 \Psi(A) s_0 = A$ proves injectivity. The term qI_4 is needed for unitality because the map $A \mapsto s_0 A t_0$ sends I_4 to pI_4 .

Set

$$U = \begin{pmatrix} s_0 I_4 & s_1 t_0 I_4 \\ 0 & t_1 I_4 \end{pmatrix}, \quad V = \begin{pmatrix} t_0 I_4 & 0 \\ s_0 t_1 I_4 & s_1 I_4 \end{pmatrix}.$$

A direct calculation using (1) gives $UV = VU = I_8$. Put

$$(10) \quad \tau = \mathrm{diag}(U, V) \in \mathrm{GL}_{16}(R).$$

The Whitehead factorization of $\mathrm{diag}(U, U^{-1})$ expresses τ as a product of elementary matrices, so $\tau \in \mathrm{EL}_{16}(R)$.

Let $H = \mathrm{EL}_{16}(R)$ and let $L = \mathrm{EL}_4(R) \leq H$ be the upper-left corner. Both groups have property (T) by the theorem of Ershov and Jaikin-Zapirain [EJZ]. Block multiplication yields

$$(11) \quad \tau \mathrm{diag}(A, I_{12}) \tau^{-1} = \mathrm{diag}(\Psi(A), I_{12}).$$

For every $i \neq j$ and $a \in R$, $\Psi(e_{ij}(a)) = e_{ij}(s_0 a t_0)$. Therefore

$$(12) \quad \tau L \tau^{-1} \leq L.$$

6. THE FULL MF RADICAL

Proposition 6.1. *The group $H = \text{EL}_{16}(R)$ is simple.*

Proof. The ring $R = L_{\mathbb{F}_2}(1, 2)$ is purely infinite simple [AA] and therefore an exchange ring [Ara]. Let $N \trianglelefteq H$. Preusser's normal-subgroup theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ such that

$$\text{EL}_{16}(R, I) \leq N \leq C_{16}(R, I),$$

where $C_{16}(R, I)$ is the full congruence subgroup of level I . Since R is simple, either $I = R$ or $I = 0$. In the first case $\text{EL}_{16}(R, I) = H$, so $N = H$. In the second case

$$N \leq C_{16}(R, 0) = Z(\text{GL}_{16}(R)).$$

The center of R is $\mathbb{F}_2$ by [AC, Corollary 4.3]. For completeness, suppose that $z \in Z(\text{GL}_{16}(R))$. Commuting z with the elementary matrices $e_{ij}(1)$ shows that $z = \lambda I_{16}$ for some $\lambda \in R$. Commuting with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times = \mathbb{F}_2^\times = \{1\}$. Thus $Z(\text{GL}_{16}(R))$ is trivial, and $N = 1$ when $I = 0$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 6.2. *Let*

$$c = e_{45}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad d^2 = 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. Every elementary generator of the upper-left 4×4 corner has both indices in $\{0, 1, 2, 3\}$. The elementary commutator relations show that it commutes with $c = e_{45}(1)$, so $c \in C_H(L)$.

Direct multiplication of the matrices in (10) gives

$$(13) \quad \tau c \tau^{-1} = e_{01}(q) e_{45}(1).$$

The second factor commutes with $\ell = e_{12}(1)$, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$. This is nontrivial because $q \neq 0$. Since R has characteristic two and $E_{02}^2 = 0$, one also has $e_{02}(q)^2 = 1$.

Let $N = \langle\langle d \rangle\rangle_H$. For arbitrary $a \in R$, two further elementary commutator identities give

$$\begin{aligned} [e_{30}(at_1), e_{02}(q)] &= e_{32}(at_1q) \in N, \\ [e_{32}(at_1q), e_{21}(s_1)] &= e_{31}(at_1qs_1) = e_{31}(a) \in N, \end{aligned}$$

where the last equality uses (8). Elementary Weyl matrices conjugate the root position $(3, 1)$ to any ordered pair of distinct indices. Thus N contains every $e_{ij}(a)$, and these elements generate H . $\square$

Proof of Theorem A. The group H is nontrivial and simple by Proposition 6.1. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ]; in particular H is finitely generated.

The containment (12), the centrality of c relative to L , and Corollary 3.3 give $d \in R_{\infty \rightarrow 2}(H)$. The subgroup $R_{\infty \rightarrow 2}(H)$ is normal, and Proposition 6.2 says that d is an involution which normally generates H . Theorem B gives $d \in \text{Rad}_{\text{MF}}(H)$. Normality of the MF radical now gives $\text{Rad}_{\text{MF}}(H) = H$.

If M is MF and $f: H \rightarrow M$ is a homomorphism, compose f with a faithful homomorphism from M to a norm matrix corona. Since $\text{Rad}_{\text{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion. $\square$

Corollary 6.3. *For every field k and every finite-dimensional k -vector space V , every homomorphism $H \rightarrow \text{GL}(V)$ is trivial.*

Proof. Let $\rho: H \rightarrow \text{GL}(V)$. The containment (12), the centrality of c relative to L , and Theorem 2.1 give $\rho(d) = 1$. Since $H = \langle\langle d \rangle\rangle_H$ by Proposition 6.2, the homomorphism ρ is trivial. $\square$

In particular, every homomorphism from H to a finite group, a residually finite group, or a compact Hausdorff group is trivial. For finite and residually finite targets this follows by the regular representation of a finite quotient; for compact targets it follows from Peter–Weyl point separation [PW].

7. PRESCRIBED MF QUOTIENTS

Let $A = \langle d \rangle \leq H$, where d is the element in Proposition 6.2. For a countable group Q , define

$$(14) \quad W_Q = H *_A (Q \times A),$$

where $A \rightarrow Q \times A$ is the inclusion into the second factor. The maps from H and $Q \times A$ to Q that are respectively trivial and the projection onto Q agree on A . They therefore define an epimorphism

$$\pi_Q: W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \rightarrow Q \times A \rightarrow W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (14) are injective; in particular, $d \neq 1$ in W_Q .

Proposition 7.1 (universal factorization). *Let T be a group such that every homomorphism $H \rightarrow T$ is trivial. Then precomposition with π_Q is a bijection*

$$\text{Hom}(Q, T) \longrightarrow \text{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Proof. Let $f: W_Q \rightarrow T$ be a homomorphism. Its restriction to H is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q, a) \in Q \times A$, we then have

$$f(q, a) = f(q, 1)f(1, a) = f(q, 1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q . The universal property of the amalgam shows that f factors through π_Q . The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d \rangle\rangle_{W_Q}$ kills H , since $\langle\langle d \rangle\rangle_H = H$, and kills the second factor A in $Q \times A$. The resulting quotient is Q , with quotient map induced by π_Q . Hence $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. $\square$

Proposition 7.2 (the hidden kernel). *Let $K_Q = \ker \pi_Q$ and, using the canonical section $Q \leq W_Q$, put $H_q = qHq^{-1}$ for $q \in Q$. Then*

$$K_Q \cong \underset{A, q \in Q}{*} H_q, \quad W_Q \cong K_Q \rtimes Q,$$

where the first group is the generalized free product of the copies H_q with their common subgroup A identified pointwise. Conjugation by $r \in Q$ sends H_q to H_{rq} and fixes A . Moreover,

$$\begin{aligned} [K_Q, K_Q] &= K_Q, & \text{Rad}_{\text{MF}}(K_Q) &= K_Q, \\ \langle\langle d \rangle\rangle_{K_Q} &= K_Q, & Z(K_Q) &= 1. \end{aligned}$$

The group K_Q is finitely generated if and only if Q is finite, and its isomorphism type depends only on the cardinality of Q .

Proof. Consider the Bass–Serre tree of the amalgam $W_Q = H *_A (Q \times A)$. Since K_Q is normal with quotient Q , the quotient graph by K_Q has one $(Q \times A)$ -type vertex, one H -type vertex for each $q \in Q$, and one edge joining the central vertex to each of the other vertices. The central vertex stabilizer is

$$K_Q \cap (Q \times A) = A,$$

the stabilizer at the vertex indexed by q is H_q , and every edge stabilizer is A . The quotient graph is a tree. Since Q commutes with A inside $Q \times A$, all its edge groups are literally the same subgroup A . The Bass–Serre normal-form theorem therefore gives the displayed generalized amalgam.

The canonical section of π_Q gives $W_Q \cong K_Q \rtimes Q$, with the stated action on the vertex groups. The group H is nonabelian and simple, hence perfect. Its conjugates H_q are therefore perfect, and they generate K_Q ; consequently K_Q is perfect.

The normal-form theorem also shows that each distinguished factor H_q is self-normalizing in K_Q , and that distinct distinguished factors are not conjugate. A central element of K_Q therefore belongs to every H_q . It is central in each such factor, while H is nonabelian simple and hence centerless. Thus $Z(K_Q) = 1$.

The common element d normally generates every vertex group H_q . Therefore its normal closure in K_Q contains every H_q , and hence $\langle\langle d \rangle\rangle_{K_Q} = K_Q$.

If Q is finite, the displayed star amalgam has finitely many finitely generated vertex groups, so K_Q is finitely generated. Conversely, every finite subset of K_Q is contained in the subamalgam supported on finitely many

indices $q \in Q$: each of its reduced normal forms involves only finitely many vertex groups. If Q is infinite, a vertex group outside this finite support contains an element of $H_q \setminus A$, and the normal form theorem shows that this element is not in the generated subamalgam. Thus K_Q is not finitely generated. Finally, the star presentation uses only the underlying indexing set of Q ; a bijection between two such sets induces an isomorphism of their kernels.

Finally, let $f: K_Q \rightarrow M$ be a homomorphism to an MF group. Its restriction to every H_q is trivial by Theorem A. Since these subgroups generate K_Q , the map f is trivial. This proves $\text{Rad}_{\text{MF}}(K_Q) = K_Q$. $\square$

Proof of Theorem C. Every homomorphism from H to an MF group is trivial by Theorem A. Proposition 7.1 therefore gives the asserted bijection for every MF group M . In particular, every homomorphism from W_Q to a norm matrix corona factors uniquely through π_Q . Intersecting their kernels gives

$$\text{Rad}_{\text{MF}}(W_Q) = \pi_Q^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d . Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\text{Rad}_{\text{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

For a normal subgroup $N \leq W_Q$, a homomorphism $f: W_Q \rightarrow M$ to an MF group kills N precisely when its unique factor $f: Q \rightarrow M$ kills $\pi_Q(N)$. Intersecting the corresponding kernels gives

$$\text{cl}_{\text{MF}}^{W_Q}(N) = \pi_Q^{-1}(\text{cl}_{\text{MF}}^Q(\pi_Q(N))).$$

The structural assertions about K_Q and the split extension are Proposition 7.2. $\square$

The same factorization applies to every target group T for which all homomorphisms $H \rightarrow T$ are trivial. Corollary 6.3 and Peter–Weyl point separation therefore give the corresponding bijections for finite-dimensional linear groups over arbitrary fields, residually finite groups, and compact Hausdorff groups.

Corollary 7.3. *Let $N \leq W_Q$. Then*

$$W_Q/N \text{ is MF} \iff \ker \pi_Q \leq N \text{ and } Q/\pi_Q(N) \text{ is MF.}$$

If every quotient of Q is MF, then

$$W_Q/N \text{ is MF} \iff d \in N.$$

In that case the stronger identity

$$\text{cl}_{\text{MF}}^{W_Q}(N) = N\langle\langle d \rangle\rangle_{W_Q}$$

holds for every normal subgroup N . In particular, $W_{\mathbb{Z}}$ is finitely generated, while its MF radical $K_{\mathbb{Z}}$ is perfect, has full MF radical, is not finitely generated, and is normally generated inside itself by d .

Proof. If W_Q/N is MF, its quotient map is a homomorphism from W_Q to an MF group. Proposition 7.1 shows that it factors through π_Q , so $\ker \pi_Q \leq N$. The induced isomorphism

$$W_Q/N \cong Q/\pi_Q(N)$$

then proves the first implication. The same isomorphism proves the converse. Suppose now that every quotient of Q is MF. The second condition in the displayed equivalence is then automatic, and $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$. Since N is normal, the condition $\ker \pi_Q \leq N$ is equivalent to $d \in N$. The MF-closure formula in Theorem C gives

$$\text{cl}_{\text{MF}}^{W_Q}(N) = \pi_Q^{-1}(\pi_Q(N)) = N \ker \pi_Q,$$

which is the asserted closure formula. Finally, every quotient of $\mathbb{Z}$ is MF. The assertions about $K_{\mathbb{Z}}$ follow from Proposition 7.2; finite generation of $W_{\mathbb{Z}}$ follows from the amalgam presentation because H , A , and $\mathbb{Z}$ are finitely generated. $\square$

Corollary 7.4 (universal endomorphisms and outer symmetry). *There is a fixed countable centerless perfect group K_{∞} such that*

$$\text{Rad}_{\text{MF}}(K_{\infty}) = K_{\infty}, \quad \langle\langle d \rangle\rangle_{K_{\infty}} = K_{\infty}, \quad \text{Sym}(\mathbb{N}) \hookrightarrow \text{Out}(K_{\infty}).$$

Here d is an involution. Consequently every countable group embeds in $\text{Out}(K_{\infty})$, and every countable monoid embeds in $\text{End}(K_{\infty})$. The group K_{∞} is both non-Hopfian and non-co-Hopfian. It also absorbs every nonempty countable amalgamated power of itself over $\langle d \rangle$. Moreover, it occurs as the characteristic MF radical in continuum many pairwise nonisomorphic countable split extensions having amenable abelian universal MF quotients.

Proof. Take

$$K_{\infty} = \ast_{A, n \in \mathbb{N}} H_n,$$

with all copies of $A = \langle d \rangle$ identified pointwise. Its stated structural properties follow from Proposition 7.2. Every self-map f of $\mathbb{N}$ induces an endomorphism Φ_f by sending H_n identically onto $H_{f(n)}$. Normal form shows that $f \mapsto \Phi_f$ is injective and detects injectivity and surjectivity of f . Thus it embeds the full transformation monoid; left regular actions give every countable monoid, while a shift and a fold show that K_{∞} is respectively non-co-Hopfian and non-Hopfian. Flattening the factor indices along $I \times \mathbb{N} \cong \mathbb{N}$ proves the asserted countable self-absorption.

Every permutation σ of $\mathbb{N}$ induces an automorphism by sending H_n identically onto $H_{\sigma(n)}$. If this automorphism were inner, then H_n and $H_{\sigma(n)}$ would be conjugate for every n . The normal-form theorem distinguishes the conjugacy classes of the distinguished factors, so $\sigma(n) = n$ for every n . This proves the injection into the outer automorphism group. Cayley's regular action embeds every countable group in $\text{Sym}(\mathbb{N})$.

For an infinite set S of primes, put

$$Q_S = \bigoplus_{p \in S} \mathbb{Z}/p\mathbb{Z}.$$

The groups Q_S are countably infinite, residually finite abelian, and hence MF [CDE]; their sets of prime element orders recover S . Theorem C gives split extensions with kernel and MF radical K_∞ and quotient Q_S . This kernel is characteristic because the MF radical is fully invariant: every endomorphism preserves the intersection of the kernels of all maps to MF groups. An isomorphism of two extensions would induce an isomorphism of their MF-radical quotients, hence of the corresponding Q_S . There are continuum many infinite sets of primes. $\square$

AI AND COMPUTATIONAL RESOURCE USAGE

AI systems were used for proof exploration, literature navigation, formal verification, and editorial revision. The authors reviewed the mathematical arguments and external references.

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